

Efficient Heat Capacity of Methane-Loaded MOF-5 with Classical Molecular Dynamics and Machine-Learned Hessians

Author name

August 25, 2026

Abstract

This report develops an efficient heat-capacity workflow for methane-loaded MOF-5 using a machine-learned interatomic potential (MLIP). Classical flexible-cell NPT molecular dynamics (MD) samples the loaded system and retains anharmonic methane and host-guest motion. Automatic differentiation (AD) through the matching PET-JAX model provides sparse Cartesian Hessians and normal-mode frequencies at optimized local minima. Following the approximation tested by Kapil *et al.*, the classical anharmonic heat capacity is corrected by the difference between quantum and classical harmonic heat capacities of the same loaded composition. The already-equilibrated empty MOF-5 structure is relaxed directly and evaluated with one reference Hessian; no empty-framework MD campaign is required. The workflow is designed to retain the main quantum correction without the cost of full-system path-integral MD, while recognizing that it is an approximation whose low-temperature and model-dependent accuracy must be validated.

1 Scientific motivation and scope

Metal-organic frameworks (MOFs) combine inorganic nodes with organic linkers to form porous crystalline networks. Their vibrational spectrum contains high-frequency local bond vibrations as well as soft linker, pore, and collective framework modes. Because the contribution of a quantum vibrational mode depends strongly on the ratio of its energy to $k_B T$, a broad frequency distribution produces a strongly temperature-dependent heat capacity. Guest adsorption adds translational, rotational, and host-guest degrees of freedom that may be weakly confined and strongly anharmonic.

The material considered here is MOF-5. Its conventional cubic cell contains 424 framework atoms with composition $C_{192}H_{96}O_{104}Zn_{32}$. In the loaded calculations, 100 methane molecules add 500 atoms, giving a 924-atom system with composition $C_{292}H_{496}O_{104}Zn_{32}$. The principal objectives are:

1. to sample the anharmonic thermodynamics of methane-loaded MOF-5 with classical NPT MD and independent replicas;
2. to compute force constants efficiently by sparse AD at optimized empty and loaded structures;
3. to obtain and converge the loaded-system harmonic quantum correction;
4. to combine that correction with the classical enthalpy derivative to estimate the loaded-system C_P efficiently; and
5. to establish a reproducible route from short cluster validation jobs to statistically meaningful production calculations.

The workflow has three computational stages. Loaded-system MD supplies the classical anharmonic enthalpy curve. Fixed-cell optimizations followed by AD Hessians supply quantum-minus-classical harmonic corrections. A final analysis combines the two terms with consistent composition and mass normalization. The empty-framework Hessian is retained as a reference and validation calculation, not as a replacement for the loaded-system correction.

2 Theory

2.1 Machine-learned interatomic potentials

2.1.1 Learning the potential-energy surface

Within the Born–Oppenheimer approximation, the electronic ground-state energy defines a potential-energy surface (PES) for the nuclei. An interatomic potential replaces an electronic-structure calculation by an approximation to this map,

$$(\mathbf{R}, \mathbf{h}, \mathbf{Z}) \mapsto U(\mathbf{R}, \mathbf{h}, \mathbf{Z}), \quad (1)$$

where $\mathbf{R} = (\mathbf{R}_1, \dots, \mathbf{R}_N)$ are the nuclear coordinates, $\mathbf{h}$ is the periodic cell, and $\mathbf{Z}$ lists the chemical species. A machine-learned interatomic potential (MLIP) represents this map by a differentiable function with parameters θ , fitted to energies and forces from electronic-structure calculations. Modern local MLIPs usually write the extensive energy as

$$U_\theta(\mathbf{R}, \mathbf{h}, \mathbf{Z}) = U_{\text{base}}(\mathbf{Z}) + \sum_{i=1}^N \varepsilon_i(\mathcal{N}_i; \theta), \quad (2)$$

where U_{base} is an optional composition-dependent baseline and $\mathcal{N}_i$ is the learned environment of atom i . The decomposition makes the cost approximately linear in N at fixed density and makes the energy size extensive. The individual atomic energies are a model construction rather than uniquely defined physical observables.

The force field is obtained by differentiating the scalar energy,

$$\mathbf{F}_i = -\frac{\partial U_\theta}{\partial \mathbf{R}_i}, \quad (3)$$

so the predicted forces are conservative, provided the same differentiable energy model is used throughout. Cell derivatives similarly give the virial or stress when the model and its interface support them. Automatic differentiation can be applied a second time to obtain the Cartesian Hessian used later in this project.

2.1.2 From an atomic structure to a graph

A graph neural network (GNN) represents the atoms by vertices and nearby interactions by directed edges. For a periodic system, an edge $(j, \mathbf{n}) \rightarrow i$ is included when

$$\mathbf{r}_{ij\mathbf{n}} = \mathbf{R}_j + \mathbf{h}\mathbf{n} - \mathbf{R}_i, \quad r_{ij\mathbf{n}} = \|\mathbf{r}_{ij\mathbf{n}}\| < r_c, \quad (4)$$

where $\mathbf{n} \in \mathbb{Z}^3$ identifies a periodic image and r_c is a cutoff. This construction removes dependence on the absolute origin. A smooth cutoff function $f_c(r)$, which tends to zero continuously at r_c , prevents discontinuous energy and force changes when a neighbor crosses the cutoff.

Neural networks operate on fixed-width numerical representations. Chemical identity is therefore mapped from a discrete species label to a trainable embedding vector,

$$\mathbf{h}_i^{(0)} = \mathbf{E}_Z[Z_i] \in \mathbb{R}^d, \quad (5)$$

where each row of the embedding matrix $\mathbf{E}_Z$ is learned during training. Geometry may be embedded through radial basis functions,

$$\mathbf{e}_{ij}^{\text{rad}} = f_c(r_{ij})[\phi_1(r_{ij}), \dots, \phi_K(r_{ij})], \quad (6)$$

and, in models that retain directional information, by the displacement $\mathbf{r}_{ij}$, its unit vector, spherical harmonics, or learned functions of these quantities. Here “embedding vector” means a vector in latent feature space; it need not transform as a physical three-dimensional vector.

2.1.3 Message passing and many-body representations

At message-passing layer ℓ , a generic GNN constructs a message on each edge, aggregates the incoming messages in a permutation-independent way, and updates the central representation:

$$\mathbf{m}_{ij}^{(\ell)} = M_\ell(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}), \quad (7)$$

$$\mathbf{m}_i^{(\ell)} = \sum_{j \in \mathcal{N}_i} \mathbf{m}_{ij}^{(\ell)}, \quad (8)$$

$$\mathbf{h}_i^{(\ell+1)} = U_\ell(\mathbf{h}_i^{(\ell)}, \mathbf{m}_i^{(\ell)}). \quad (9)$$

The learned maps M_ℓ and U_ℓ are commonly multilayer perceptrons or attention blocks. Although the graph initially contains only pairwise edges, repeated updates create many-body features: after L layers, an atom can depend on paths extending up to L graph hops. A readout map then converts the final features into the local energies in Eq. (2).

The Point Edge Transformer (PET) used by PET-MAD is a particular GNN that stores a token $\mathbf{f}_{ij}^{(\ell)} \in \mathbb{R}^{d_{\text{PET}}}$ on every directed edge rather than storing information only on atoms [4, 3]. For the environment centered on atom i , PET encodes the neighbor species, relative Cartesian position, and incoming edge message into a token of the schematic form

$$\mathbf{p}_{ij}^{(\ell)} = \sigma(\mathbf{W}_r^{(\ell)} \mathbf{r}_{ij}), \quad (10)$$

$$\mathbf{t}_{ij}^{(\ell)} = \text{MLP}_\ell(\mathbf{f}_{ji}^{(\ell-1)} \parallel \mathbf{E}_Z^{(\ell)}[Z_j] \parallel \mathbf{p}_{ij}^{(\ell)}), \quad (11)$$

where $\parallel$ denotes concatenation and σ is a nonlinear activation. In the first layer the incoming message is absent. A transformer mixes all tokens in one atomic environment by self-attention. For one attention head,

$$\mathbf{q}_a = \mathbf{W}_Q \mathbf{t}_a, \quad \mathbf{k}_b = \mathbf{W}_K \mathbf{t}_b, \quad \mathbf{v}_b = \mathbf{W}_V \mathbf{t}_b, \quad (12)$$

$$\alpha_{ab} = \frac{\exp(\mathbf{q}_a^T \mathbf{k}_b / \sqrt{d_k}) f_c(r_{ib})}{\sum_c \exp(\mathbf{q}_a^T \mathbf{k}_c / \sqrt{d_k}) f_c(r_{ic})}, \quad \mathbf{y}_a = \sum_b \alpha_{ab} \mathbf{v}_b. \quad (13)$$

Multi-head attention repeats this operation in several learned subspaces, followed by feed-forward layers and residual connections. Because no ordering of the neighbors enters Eq. (13), permuting the atoms permutes the edge outputs consistently. PET interprets these outputs as the messages for the next GNN layer and applies learned readout heads across edges and layers to form the total energy.

2.1.4 Physical symmetries

For an isolated non-magnetic system without external fields, a valid scalar energy should be unchanged by translation, rotation or reflection of the whole structure and by permutation of identical atoms:

$$U(\mathbf{R}) = U(\mathbf{R} + \mathbf{a}) = U(\mathbf{Q}\mathbf{R}) = U(\mathbf{P}\mathbf{R}), \quad \mathbf{Q} \in O(3). \quad (14)$$

An invariant scalar is unchanged by a symmetry operation, whereas an equivariant vector changes in the corresponding way; for example, $\mathbf{F}_i(\mathbf{Q}\mathbf{R}) = \mathbf{Q}\mathbf{F}_i(\mathbf{R})$. Translation invariance follows from using relative positions, and the shared local functions and symmetric aggregation enforce permutation invariance. Many MLIPs impose rotational symmetry exactly through invariant geometric descriptors or $E(3)$ -equivariant features. PET instead processes Cartesian displacement vectors with an unconstrained transformer and learns rotational behavior from random rotational augmentation [4, 3]. Thus the PET-MAD model used here is not intrinsically exactly rotation invariant, although its reported rotational discrepancies are small compared with its prediction errors.

2.1.5 Training and domain of validity

Given reference configurations s , a typical energy–force training objective is

$$\mathcal{L}(\theta) = \sum_s \left[w_E \left(\frac{U_\theta^{(s)} - U_{\text{ref}}^{(s)}}{N_s} \right)^2 + \frac{w_F}{3N_s} \sum_{i\alpha} \left(F_{i\alpha,\theta}^{(s)} - F_{i\alpha,\text{ref}}^{(s)} \right)^2 \right], \quad (15)$$

possibly augmented by stress terms and regularization. Because the force labels are derivatives of the PES, each configuration supplies $3N_s$ local slope constraints in addition to its energy. Optimization adjusts the embeddings, message-passing transformations and readout weights to minimize this loss. PET-MAD was trained on a chemically diverse DFT dataset rather than on MOF-5 alone, making it a universal pretrained potential [3].

An MLIP interpolates the reference PES represented by its training data; it does not replace the approximations of the underlying electronic-structure method and need not extrapolate reliably to unfamiliar bonding, charge states, very short distances or thermodynamic phases. Energy and force test errors, trajectory stability and uncertainty indicators are therefore necessary but distinct checks. This project also uses second derivatives, which can be more sensitive than energies or forces and were not directly constrained by the usual loss in Eq. (15). Curvatures and vibrational spectra must therefore be validated explicitly.

2.2 Classical molecular dynamics

2.2.1 Hamiltonian dynamics and statistical averages

Molecular dynamics (MD) generates a trajectory through phase space by propagating the nuclear positions and momenta on a chosen PES. For classical point nuclei with masses m_i , the Hamiltonian is

$$\mathcal{H}(\mathbf{R}, \mathbf{P}) = \sum_{i=1}^N \frac{\mathbf{P}_i^2}{2m_i} + U(\mathbf{R}), \quad (16)$$

and Hamilton’s equations give

$$\dot{\mathbf{R}}_i = \frac{\mathbf{P}_i}{m_i}, \quad \dot{\mathbf{P}}_i = -\frac{\partial U}{\partial \mathbf{R}_i} = \mathbf{F}_i. \quad (17)$$

In exact microcanonical dynamics, $\mathcal{H}$ is conserved. If the dynamics is ergodic and the trajectory is sufficiently long, an equilibrium average can be estimated by a time average,

$$\langle A \rangle = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A(\mathbf{R}(t), \mathbf{P}(t)) dt. \quad (18)$$

Finite trajectories contain correlated samples, so their statistical precision is controlled by the simulation length relative to the integrated autocorrelation time, not simply by the number of saved frames.

The continuous equations are replaced by a finite-timestep integrator. A representative symplectic scheme is velocity Verlet,

$$\mathbf{P}_i^{n+1/2} = \mathbf{P}_i^n + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^n), \quad (19)$$

$$\mathbf{R}_i^{n+1} = \mathbf{R}_i^n + \Delta t \frac{\mathbf{P}_i^{n+1/2}}{m_i}, \quad (20)$$

$$\mathbf{P}_i^{n+1} = \mathbf{P}_i^{n+1/2} + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^{n+1}). \quad (21)$$

It is time reversible and symplectic, and has global error of order $\mathcal{O}(\Delta t^2)$. It approximately conserves a nearby “shadow” Hamiltonian rather than the exact one. The timestep must resolve the fastest vibration; an apparently controlled temperature does not by itself rule out integration error.

2.2.2 Temperature, ensembles and thermostats

For g unconstrained kinetic degrees of freedom, the instantaneous kinetic temperature is conventionally defined through equipartition,

$$T_{\text{kin}} = \frac{2K}{gk_{\text{B}}}, \quad K = \sum_i \frac{\mathbf{P}_i^2}{2m_i}. \quad (22)$$

Temperature is an ensemble property; T_{kin} fluctuates along a finite canonical trajectory. Constant-energy (NVE) dynamics samples the microcanonical ensemble. To sample the canonical ensemble, whose stationary phase-space density is

$$\rho_{NVT}(\mathbf{R}, \mathbf{P}) \propto \exp[-\beta \mathcal{H}(\mathbf{R}, \mathbf{P})], \quad \beta = (k_{\text{B}}T)^{-1}, \quad (23)$$

the physical variables may be coupled to a thermostat.

For example, Langevin dynamics adds friction and random forces,

$$m_i \ddot{\mathbf{R}}_i = \mathbf{F}_i - m_i \gamma \dot{\mathbf{R}}_i + \boldsymbol{\eta}_i(t), \quad (24)$$

$$\langle \eta_{i\alpha}(t) \eta_{j\beta}(t') \rangle = 2m_i \gamma k_{\text{B}} T \delta_{ij} \delta_{\alpha\beta} \delta(t - t'). \quad (25)$$

The second relation is the fluctuation–dissipation condition that makes the canonical distribution stationary. Weak friction perturbs the natural dynamics less but exchanges heat slowly; strong friction controls temperature rapidly but can overdamp motion. Nosé–Hoover-chain thermostats instead add a sequence of deterministic extended variables and are designed to generate the same canonical distribution while improving ergodicity over a single Nosé–Hoover variable [8]. Thermostat choice affects sampling efficiency and dynamical correlation functions even when equilibrium averages are unchanged.

An isothermal–isobaric simulation additionally evolves the volume or cell so that the target density is proportional to $\exp[-\beta(\mathcal{H} + PV)]$. Flexible-cell algorithms must include the appropriate cell kinetic variables, Jacobian and virial terms; the Martyna–Tobias–Klein scheme is one such extended-system formulation [7]. Reliable NPT dynamics therefore requires a potential with validated cell derivatives or stresses. NVE, NVT and NPT trajectories answer different thermodynamic questions and should not be interchanged solely because their mean temperatures are similar.

2.3 Path-integral molecular dynamics

2.3.1 Quantum statistical mechanics as a ring polymer

Classical MD neglects the wave-like character of the nuclei. For distinguishable quantum nuclei on a single Born–Oppenheimer surface, the Hamiltonian operator and canonical partition

function are

$$\hat{\mathcal{H}} = \sum_i \frac{\hat{\mathbf{P}}_i^2}{2m_i} + U(\hat{\mathbf{R}}), \quad Z = \text{Tr} \left[e^{-\beta \hat{\mathcal{H}}} \right]. \quad (26)$$

The kinetic and potential operators do not commute. Splitting the imaginary time $\beta\hbar$ into P slices with a Trotter factorization maps the quantum partition function, in the limit $P \rightarrow \infty$, onto a classical extended system. Its primitive ring-polymer Hamiltonian is

$$Z_P \propto \int d\mathbf{R} d\mathbf{P} e^{-\beta_P H_P}, \quad \beta_P = \frac{\beta}{P}, \quad (27)$$

$$H_P = \sum_{s=1}^P \left[\sum_i \frac{|\mathbf{P}_i^{(s)}|^2}{2m_i} + \sum_i \frac{m_i \omega_P^2}{2} |\mathbf{R}_i^{(s)} - \mathbf{R}_i^{(s+1)}|^2 + U(\mathbf{R}^{(s)}) \right], \quad \omega_P = \frac{P}{\beta\hbar}, \quad (28)$$

with cyclic boundary condition $\mathbf{R}_i^{(P+1)} = \mathbf{R}_i^{(1)}$ [6]. Each physical atom is thus represented by P replicas, or beads, joined by harmonic springs. The physical potential is evaluated independently on every bead, while the springs encode the quantum delocalization of the nuclei in imaginary time. When $P = 1$, Eq. (28) reduces to the classical configurational distribution. Larger P is required as temperature falls or as the maximum physical vibrational frequency increases; convergence must be checked for the chosen isotope masses and temperature range.

Two useful geometrical quantities are the bead centroid and radius of gyration,

$$\bar{\mathbf{R}}_i = \frac{1}{P} \sum_{s=1}^P \mathbf{R}_i^{(s)}, \quad r_{g,i}^2 = \frac{1}{P} \sum_{s=1}^P |\mathbf{R}_i^{(s)} - \bar{\mathbf{R}}_i|^2. \quad (29)$$

The centroid describes the mean imaginary-time position, while the spread of the beads measures nuclear quantum delocalization. For a position-dependent operator $A(\mathbf{R})$, the primitive estimator has the form

$$\langle A \rangle = \lim_{P \rightarrow \infty} \left\langle \frac{1}{P} \sum_{s=1}^P A(\mathbf{R}^{(s)}) \right\rangle_{H_P}. \quad (30)$$

Kinetic energy, pressure and heat capacity require specialized estimators or thermodynamic derivatives; they are not obtained by treating the fictitious ring-polymer kinetic energy as the physical nuclear kinetic energy.

2.3.2 Sampling the ring polymer

PIMD introduces fictitious momenta in order to sample the ring-polymer distribution using MD algorithms. A discrete Fourier or normal-mode transformation diagonalizes the free ring-polymer spring term. Its mode frequencies are

$$\omega_k = 2\omega_P \sin \left(\frac{k\pi}{P} \right), \quad k = 0, \dots, P-1, \quad (31)$$

where $k = 0$ is the centroid mode. The large separation between the slow physical motion and the stiff internal modes makes naive sampling inefficient. Normal-mode or staging coordinates, mass preconditioning, multiple time steps, and mode-specific thermostats address this stiffness. The PILE thermostat applies analytically tuned Langevin friction to the known free ring-polymer modes, while the PILE-L variant also thermostats the centroid [9].

The standard primitive factorization has an asymptotic discretization error of order $\mathcal{O}(P^{-2})$. The fourth-order Suzuki–Chin factorization used in the reference work replaces the bead potential by a bead-dependent effective potential of the general form

$$\tilde{U}_s(\mathbf{R}; \beta) = w_s U(\mathbf{R}) + w_s d_s \left(\frac{\beta\hbar}{P} \right)^2 \sum_i \frac{|\mathbf{F}_i(\mathbf{R})|^2}{m_i}, \quad (32)$$

where w_s and d_s are factorization coefficients that alternate between beads. The force-squared correction accounts for the leading non-commutation of kinetic and potential operators and gives fourth-order convergence for appropriate thermodynamic observables. Direct differentiation would involve Hessians; finite-difference implementations make the scheme practical with general force providers [10]. Ring-polymer contraction can further reduce cost by evaluating slowly varying, long-range components of the potential on fewer contracted beads, while rapidly varying short-range terms remain evaluated on the full polymer.

2.3.3 What PIMD does and does not predict

The trajectories generated by ordinary PIMD are a sampling device for static quantum equilibrium averages. The bead index represents imaginary time, and the artificial bead masses, thermostats and ring-polymer vibrations have no general identification with real nuclear motion. Consequently, ordinary PIMD does not directly yield quantum time-correlation functions, diffusion coefficients or vibrational spectra. Ring-polymer molecular dynamics (RPMD) and centroid molecular dynamics are related but additional approximations for real-time correlation functions; they should not be conflated with PIMD.

PIMD captures nuclear zero-point fluctuations, tunnelling contributions to equilibrium statistics, and anharmonic sampling on the supplied PES, subject to convergence in bead count, timestep, sampling length and system size. It does not correct errors in that PES, include electronic excited states, or automatically change an NVT calculation into a constant-pressure one. For the heat capacity in the reference study, the ring polymer was sampled in a constant-pressure ensemble and C_P was obtained from converged enthalpy derivatives at neighboring temperatures, as discussed below.

2.4 Thermodynamic definitions

2.4.1 Constant-volume and constant-pressure heat capacities

For a system at equilibrium, the constant-volume and constant-pressure heat capacities are

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,N}, \quad (33)$$

$$C_P = \left(\frac{\partial \langle H \rangle}{\partial T} \right)_{P,N}, \quad H = E + PV, \quad (34)$$

where E is the total internal energy and H is the enthalpy. For a stable single phase,

$$C_P - C_V = \frac{TV\alpha_V^2}{\kappa_T}, \quad (35)$$

where $\alpha_V = V^{-1}(\partial V/\partial T)_P$ is the volumetric thermal expansion coefficient and $\kappa_T = -V^{-1}(\partial V/\partial P)_T$ is the isothermal compressibility. Equation (35) explains why C_V and C_P need not be interchangeable even when their numerical difference is small for a particular solid.

In an equilibrated classical canonical ensemble, C_V can also be obtained from total-energy fluctuations,

$$C_V = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}. \quad (36)$$

The analogous isothermal–isobaric expression uses enthalpy fluctuations. These relations require a stationary ensemble, long sampling, and uncertainty estimation that accounts for time correlation. Moreover, classical nuclear dynamics gives equipartition rather than the quantum freezing of high-frequency modes. The present workflow obtains its classical constant-pressure term by differentiating replica-averaged NPT enthalpies on a temperature grid. It obtains the quantum correction separately from harmonic normal-mode frequencies.

2.4.2 Why classical and quantum predictions differ

In the classical harmonic limit, every quadratic vibrational degree of freedom contributes k_B to C_V , producing the Dulong–Petit limit at all temperatures. Quantum mechanically, a mode of angular frequency ω becomes thermally populated only when $\hbar\omega$ is comparable to or smaller than $k_B T$. High-frequency framework and molecular modes are therefore partially frozen near room temperature. Classical MD can still describe anharmonic structural motion, but it cannot generate the correct quantum populations simply by extending the trajectory.

3 Reference study: Kapil *et al.*

Kapil *et al.* studied empty MOF-5 and cells containing 50, 100, and 150 methane molecules using classical MD, Suzuki–Chin path-integral molecular dynamics (PIMD), and quantum harmonic normal-mode analysis [1]. Their work provides both the physical motivation and an important limit on what the present calculation can claim.

3.1 Force field and starting configurations

The reference work used analytical QuickFF force fields derived from electronic-structure calculations, not PET-MAD [5]. Framework covalent terms were fitted to B3LYP cluster calculations, with MBIS charges obtained from a periodic PBE density. Nonbonded interactions combined Gaussian-distributed electrostatics with an MM3 Buckingham model and a 15 Å cutoff with tail corrections. Methane was inserted with RASPA, configurations with unrealistic contacts were rejected, and canonical Monte Carlo was used before dynamics. Consequently, agreement of timestep and temperature alone would not make the present PET-MAD simulations a numerical reproduction of the paper.

3.2 Published simulation protocols

Parameter	Classical MD	Suzuki–Chin PIMD
State points	100–500 K, 1 bar	100–500 K, 1 bar
Cell dynamics	Fully flexible cell, zero deviatoric stress	Fluctuating volume with cubic cell shape retained
Integrator	Verlet, 0.5 fs	BAOAB-type multiple-time-step scheme
Thermostat	Three-element Nosé–Hoover chain, 100 fs relaxation	PILE-L ring-polymer thermostat; 100 fs relaxation
Barostat	Martyna–Tobias–Klein, 1000 fs relaxation	Bussi–Zykova–Parrinello form adapted to Suzuki–Chin; 1000 fs relaxation
Quantum discretization	Not applicable	64 beads; long-range forces contracted to 8 beads
Force timesteps	0.5 fs	0.25 fs short range; 1 fs long range
Empty MOF-5 sampling	One 500 ps trajectory	Five independent 125 ps trajectories
Loaded sampling	Five independent 500 ps trajectories	Thirty independent 50 ps trajectories
Discarded equilibration	First 100 ps	First 25 ps
Heat-capacity estimator	Classical thermodynamic sampling	Centered enthalpy derivative with $\Delta T = 25$ K

Table 1: Principal simulation parameters reported by Kapil *et al.* Numerical random seeds and output strides were not reported.

The associated Nosé–Hoover-chain, constant-pressure, path-integral, and PILE-L methods are described in Refs. [8, 7, 6, 9]. Their production C_P curves were evaluated from neighboring enthalpies,

$$C_P(T) \simeq \frac{\mathcal{H}(T + \Delta T) - \mathcal{H}(T - \Delta T)}{2\Delta T}, \quad \Delta T = 25 \text{ K}, \quad (37)$$

rather than from a single short trajectory. Every temperature point therefore depends on converged, statistically independent simulations at adjacent temperatures.

For the harmonic comparison, optimized structures were used to construct Cartesian Hessians with Yaff and normal modes with TAMkin. Empty MOF-5 was tested with both QuickFF and UFF4MOF. The loaded 100-methane calculation used five different Monte Carlo configurations, each optimized into a local minimum before its harmonic spectrum was evaluated.

3.3 Physical conclusions relevant to this project

The main findings of the reference study are:

- Empty MOF-5 is quantum mechanical but nearly harmonic over the studied range. Its quantum harmonic C_V is close to the PIMD C_P , with a room-temperature gravimetric value of approximately 0.7 J/g/K.
- Classical MD overestimates the empty-framework heat capacity because it populates all vibrational modes according to equipartition.
- Methane loading introduces strongly anharmonic host–guest motion. The PIMD curves develop a loading-dependent minimum around 200 K, caused mainly by changes in attractive host–guest interactions as confined methane becomes more mobile.
- A harmonic calculation cannot represent diffusion, cage-to-cage motion, or the temperature-dependent localization of methane. Nuclear quantum effects and anharmonic sampling are both needed for a quantitative loaded-system C_P .
- The approximate combination

$$C \approx C_{\text{qn}}^{\text{har}} - C_{\text{cl}}^{\text{har}} + C_{\text{cl}}^{\text{anh}} \quad (38)$$

qualitatively recovers the loaded minimum and becomes useful above roughly 200 K in their 100-methane test. This is the approximation implemented in the current repository.

4 Present computational methodology

4.1 Structure preparation and methane insertion

The repository host structure is the conventional cubic MOF-5 cell with 424 atoms and a cell parameter of approximately 25.866 Å. The loaded campaign is generated by `python -m mof_heat_capacity.protocols.loaded`. For every temperature and replica, methane molecules are centered, uniformly randomly rotated using a quaternion, and placed at uniformly sampled fractional coordinates. Periodic atom–atom distances are checked against all previously placed atoms. A 100-methane structure contains 924 atoms. Independent insertion and velocity seeds prevent the nominal replicas from sharing an identical initial state.

This procedure prevents severe initial overlaps, but it is not an adsorption equilibrium calculation. In particular, it does not reproduce the RASPA insertion and Monte Carlo equilibration used by Kapil *et al.* Several independent insertion seeds and adequate subsequent equilibration are required before drawing conclusions about loading-dependent thermodynamics.

4.2 PET-MAD energy model

The general construction of a local MLIP and the PET architecture are given in Section 2. For the checkpoint used in this project, the model metadata specifies an 8 Å cutoff, three GNN layers, and an adaptive maximum of 40 neighbors. At inference, the learned parameters θ are fixed and Eqs. (2) and (3) are evaluated for each MD configuration.

MD loads a metatomic PET-MAD or PET-SOL export. Harmonic analysis converts the matching checkpoint to a PET-JAX representation. Using the same learned parameters in both paths is essential: a trajectory from one potential must not be differentiated with a different potential. The updated exports expose an explicit stress output, and loaded-campaign configurations record stress validation explicitly rather than inferring capability from a model filename. PET is a universal model rather than a MOF-5-specific fit, so energy, force, stress, and especially curvature accuracy for the relevant structures must still be documented and validated [3].

4.3 Classical NPT molecular dynamics

The statistical and dynamical foundations of classical MD are summarized in Section 2.2. Only methane-loaded MOF-5 is propagated. LAMMPS evaluates a metatomic PET model in a flexible triclinic cell at an isotropic target pressure of 1 bar. Temperature and pressure are controlled by a Nosé–Hoover-chain/MTTK NPT fix, with chain length 3, thermostat damping time 100 fs, and barostat damping time 1000 fs. The production template specifies 10^6 steps of 0.5 fs, hence 500 ps, saves data every 100 steps (0.05 ps), and discards the first 100 ps. The initial campaign uses one replica at 100, 200, 300, 400, and 500 K. This coarse 100 K spacing reduces cost but limits resolution of curvature in the enthalpy curve and of the heat-capacity minimum; denser temperatures and independent replicas remain convergence extensions.

The exported model must expose stress and must be validated for the intended MOF/loading states before NPT use. Mean loaded-system enthalpies are computed from the stationary portions of the trajectories,

$$\langle H \rangle_T = \langle E + P_{\text{ext}} V \rangle_T, \quad (39)$$

then averaged across replicas and differentiated with respect to temperature. This classical contribution retains guest diffusion, localization changes, and other anharmonic effects sampled by the MLIP.

4.4 Harmonic approximation and Cartesian Hessian

Let $\mathbf{R}_0$ be a stationary reference structure and $\mathbf{u} = \mathbf{R} - \mathbf{R}_0$. Expanding the potential energy gives

$$U(\mathbf{R}_0 + \mathbf{u}) = U(\mathbf{R}_0) + \underbrace{\nabla U(\mathbf{R}_0) \cdot \mathbf{u}}_{=0 \text{ at a minimum}} + \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} + \mathcal{O}(u^3), \quad (40)$$

with Cartesian Hessian

$$H_{i\alpha, j\beta} = \frac{\partial^2 U}{\partial R_{i\alpha} \partial R_{j\beta}} = - \frac{\partial F_{i\alpha}}{\partial R_{j\beta}}. \quad (41)$$

For N atoms, $\mathbf{H}$ has dimensions $3N \times 3N$; for the 100-methane cell this is 2772×2772 .

PET-JAX differentiates the model computation graph rather than using displaced finite differences. The implementation uses forward-over-reverse Hessian–vector products,

$$\mathbf{H} \mathbf{v} = \left. \frac{d}{d\epsilon} \nabla U(\mathbf{R} + \epsilon \mathbf{v}) \right|_{\epsilon=0}. \quad (42)$$

Because a local message-passing model couples only graph-connected atoms, the Hessian has a block-sparse structure. SADMOF [11] constructs a graph-derived sparsity pattern and ASDEX

colors its columns so that several independent columns can be reconstructed from one differentiated evaluation. The current settings are single precision, graph depth `hops=3`, HVP chunk size 1, and rematerialization enabled. These are computational choices, not established convergence values. In particular, the graph-hop pattern must be enlarged until frequencies and C_V stop changing, and should be checked against a dense or less aggressively truncated reference where feasible.

4.5 Normal modes and quantum harmonic heat capacity

The Hessian is symmetrized and mass weighted to obtain

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}}, \quad D\mathbf{e}_k = \omega_k^2 \mathbf{e}_k. \quad (43)$$

The code enforces the force-constant acoustic sum rule before diagonalization,

$$\sum_j H_{i\alpha,j\beta} = 0, \quad (44)$$

so that uniform translations have zero restoring force. The upstream SADMOF observable maps negative eigenvalues to zero frequency for compatibility with its reference heat-capacity convention. The present workflow instead diagonalizes the mass-weighted Hessian locally and retains unstable eigenvalues as signed negative frequencies. Hybrid assembly rejects unexpected imaginary modes and permits at most the three translational near-zero modes, so a thermally displaced frame or saddle point cannot silently enter the correction.

For each retained mode, define

$$x_k = \frac{\hbar\omega_k}{k_B T} = \frac{hc\tilde{\nu}_k}{k_B T}, \quad (45)$$

where $\tilde{\nu}_k$ is the wavenumber. The quantum harmonic contribution is

$$C_{V,k} = k_B \left[\frac{x_k/2}{\sinh(x_k/2)} \right]^2, \quad (46)$$

and the gravimetric result is

$$c_V(T) = \frac{N_A}{M_{\text{cell}}} \sum_{k \in \mathcal{V}} C_{V,k}, \quad (47)$$

in J/g/K. The mass M_{cell} includes methane for a loaded system, so both the spectrum and the normalization change with loading.

One Hessian produces the entire configured 100–500 K curve in 10 K increments. The temperature appearing in Eq. (47) controls oscillator populations; it does not require a separate MD trajectory at every analysis temperature when the force constants are assumed temperature independent.

4.6 Optimized minima and the hybrid correction

The empty-framework calculation starts from the equilibrated MOF-5 structure already available to the project. Its coordinates are relaxed at fixed cell with the selected MLIP and one AD Hessian is evaluated. No empty-MOF MD is needed merely to obtain a Hessian reference.

For the loaded system, final structures from several independent classical MD replicas are quenched at fixed cell with the same MLIP. A Hessian is accepted only after the relaxation reaches the chosen force tolerance and its signed spectrum is checked for unexpected imaginary

modes. Raw finite-temperature frames are not treated as normal-mode reference structures. If N_m independent optimized minima are retained, the mean harmonic quantum correction is

$$\overline{\Delta c^{\text{har}}}(T) = \frac{1}{N_m} \sum_{s=1}^{N_m} \left[c_{\text{qn},s}^{\text{har}}(T) - c_{\text{cl},s}^{\text{har}} \right], \quad (48)$$

with sample variance

$$s_{\Delta c}^2(T) = \frac{1}{N_m - 1} \sum_{s=1}^{N_m} \left[\Delta c_s^{\text{har}}(T) - \overline{\Delta c^{\text{har}}}(T) \right]^2. \quad (49)$$

For each minimum, the quantum and classical harmonic terms use exactly the same retained-mode mask. The classical term is k_B per retained mode. The complete loaded-system spectrum is used because framework and methane motion can mix; the empty Hessian alone omits methane quantization and cannot supply this correction.

The final estimator is

$$c_P^{\text{approx}}(T) = c_{P,\text{cl}}^{\text{anh}}(T) + \overline{\Delta c^{\text{har}}}(T), \quad c_{P,\text{cl}}^{\text{anh}}(T) = \frac{d\langle H \rangle_{\text{cl}}}{dT}. \quad (50)$$

Extensive terms are combined before division by the total mass of the loaded cell. Replica-aware resampling propagates uncertainty in the enthalpy curve; the spread across optimized minima measures sensitivity of the harmonic correction. The Hessian contribution is formally a C_V correction added to a classical C_P term, following the small solid-state $C_P - C_V$ assumption used in the reference approximation. This mixing and the approximation’s limited validation below roughly 200 K must be stated with the result.

5 Results and discussion

The earlier trajectory and raw-frame Hessian calculations were exploratory and are not results of the focused workflow defined here. Their numerical tables and generated overview figure have therefore been removed from this report. Production results should be added only after the new loaded replicas, optimized-minimum Hessians, and hybrid analysis have completed and passed the checks below.

The primary reported result will be a temperature-resolved table or curve containing the replica-averaged classical enthalpy derivative, the mean loaded harmonic quantum correction, their separate uncertainties, and the combined c_P^{approx} . The empty quantum-harmonic curve will be shown as a separately normalized reference. Density, cell response, framework stability, methane mobility, imaginary-mode counts, and relaxation residual forces will support interpretation rather than replace the heat-capacity result.

6 Recommended production and analysis protocol

6.1 Empty MOF-5 reference

The equilibrated empty-framework structure is used directly; an empty-MOF MD campaign would add cost without supplying the anharmonic loaded-system term needed by Eq. (50).

1. Relax the atomic coordinates with the same PET-MAD checkpoint used for the Hessian. Report maximum residual force and do not relax the cell unless a stress-capable, validated model is adopted.
2. Check stability using signed eigenvalues and inspect low-frequency modes. A large number of unstable modes invalidates the harmonic interpretation.

3. Converge float32 against float64, graph hops, HVP chunking, acoustic sum-rule treatment, and frequency threshold.
4. Converge the conventional-cell result with respect to supercell size or phonon wave-vector sampling. A single periodic cell Hessian samples only the corresponding finite set of modes.
5. Retain the resulting curve as a separately normalized reference and compare its room-temperature value with the approximately 0.7 J/g/K reference scale, while stating that PET-MAD replaces QuickFF and that model error remains to be quantified.

6.2 Loaded MOF-5 production and Hessians

For the 100-methane system, the production sequence is:

1. Prepare independent methane placements and velocity seeds for each replica on the enthalpy temperature grid.
2. Run flexible-cell classical NPT only for the loaded system. Determine equilibration from enthalpy, cell, framework, and methane-mobility diagnostics rather than assuming the template duration is sufficient.
3. Average enthalpy over the stationary part of each replica, combine replica uncertainties, and differentiate the temperature dependence.
4. Select representative independent loaded configurations, quench each one to a fixed-cell minimum with the same MLIP, and reject unconverged or unstable minima.
5. Compute one AD Hessian per accepted minimum. Converge precision, graph hops, HVP chunking, acoustic treatment, frequency threshold, and cell size on the quantum-minus-classical correction.
6. Average the loaded corrections, combine them with the classical enthalpy derivative using Eq. (50), and report MD and minimum-to-minimum uncertainties separately as well as jointly.

This calculation is intentionally not an exact reproduction of the reference study. A small number of path-integral state points may be considered later as validation if quantitative accuracy below approximately 200 K is required, but they are not part of the production workflow.

6.3 Equilibration and convergence criteria

Production decisions should be based on observables, not a fixed waiting time alone. At minimum, inspect:

- running and block averages of temperature, potential energy, and total energy;
- maximum and distribution of force magnitudes;
- framework bond integrity, pore geometry, and cell consistency;
- methane center-of-mass distributions, host-guest distances, and mean-squared displacements;
- sensitivity to timestep and thermostat friction;
- agreement among independent seeds;

- convergence of the enthalpy derivative with replica count, sampling length, and temperature spacing; and
- convergence of the harmonic correction with independent minima, precision, sparsity depth, cell size, and mode treatment.

The timestep should first be validated in short NVE tests by checking energy drift. Thermostat convergence should then be tested separately, because a thermostat can conceal an integration error while maintaining the desired kinetic temperature.

7 Implementation, outputs, and reproducibility

The production campaign is prepared by `python -m mof_heat_capacity.protocols.loaded` and submitted by `simulation/submit_loaded_md.sh`. These commands create independent loaded structures and LAMMPS/metatomic NPT runs while deliberately rejecting a zero-methane MD campaign. Trajectory diagnostics are submitted with `properties/submit_analysis.sh`.

The command `properties/submit_heat_capacity.sh` takes final structures from selected loaded replicas, relaxes them with `mof_heat_capacity.structures.relax`, and passes the optimized structures to `mof_heat_capacity.analysis.harmonic`. The same command relaxes and evaluates the supplied empty reference directly. Harmonic archives store the run and model identity, optimized structure, temperature grid, frequency array, gravimetric C_V curve, precision, graph hops, HVP chunk size, and rematerialization choice. Finally, `properties/submit_hybrid_analysis.sh` differentiates the loaded NPT enthalpies and writes the classical term, harmonic correction, combined curve, uncertainties, and provenance as NPZ, CSV, and JSON files.

Cluster calculations use one node, one task, and one GPU because neither the MD nor Hessian code is MPI or multi-GPU parallel. Independent temperature, replica, or optimized-minimum jobs are the available level of parallelism. Slurm job identifiers and resource use should be retained; elapsed time and peak memory can be recovered with, for example,

```
sacct -j JOB_ID --format=JobID,State,Elapsed,MaxRSS,AllocTRES
```

Wall time and memory requests should be based on representative jobs. Separate output directories and prefixes are required for every loading, temperature, seed, and replica.

A reproducible result should record:

- source structure, cell, atom count, chemical composition, loading, and methane-insertion seed and rejection criteria;
- PET-MAD checkpoint hash, metatomic export, PET-JAX conversion, software environment, device, and numerical precision;
- ensemble, target temperature, timestep, thermostat and friction, number of steps, output stride, velocity seed, equilibration interval, production interval, and restart history;
- Hessian source replica, relaxation state, residual forces, graph hops, HVP chunk size, rematerialization, acoustic sum rule, frequency threshold, and temperature grid; and
- independent-replica count, autocorrelation or block length, uncertainty estimator, and all convergence tests.

8 Limitations and interpretation

The principal limitations of the planned result are:

- PET-MAD and QuickFF describe different potential-energy surfaces; agreement with the paper cannot be assumed.
- Flexible-cell NPT requires a stress-capable MLIP and a MOF-5-specific stress benchmark; an output capability flag alone is not validation.
- Classical nuclei are used in MD; nuclear quantum effects enter only through the harmonic oscillator formula.
- The random insertion is not an adsorption or Monte Carlo equilibrium distribution.
- A finite number of quenched structures may not represent every thermodynamically important loaded basin. Their spread measures sensitivity but is not automatically a free-energy-weighted average.
- The harmonic correction cannot itself represent methane diffusion or barrier crossing; those effects enter only through classical MD.
- Adding a harmonic C_V correction to a classical C_P term assumes that the corresponding pressure–volume quantum correction is small.
- The approximation was quantitatively successful in the reference 100-methane system mainly above about 200 K; transfer to this MLIP, other loadings, and lower temperatures requires validation.
- Finite-size, sparse-Hessian depth, precision, temperature-derivative, model-validity, and independent-replica errors must all be quantified.

These qualifications define the boundary between an efficient, physically motivated approximation and an exact quantum-anharmonic thermodynamic result. They should be carried into every plot, table, and comparison with experiment.

9 Conclusions

The focused workflow avoids both unnecessary empty-framework MD and the cost of a full production path-integral campaign. Classical NPT trajectories are reserved for loaded MOF-5, where they are needed to describe anharmonic guest and host–guest motion. Sparse AD Hessians are evaluated only at converged fixed-cell minima and provide the loaded-system quantum-minus-classical harmonic correction. The equilibrated empty structure contributes one direct reference Hessian.

The final quantity is the classical loaded-system enthalpy derivative plus the mean loaded harmonic correction. Its credibility depends on stress validation, stationary and replicated MD, stable optimized minima, numerical Hessian convergence, consistent mode and mass bookkeeping, and explicit uncertainty propagation. This strategy implements the cost-saving approximation motivated by Kapil *et al.* without attempting to reproduce every calculation in their study.

References

- [1] V. Kapil *et al.*, *Modeling the Structural and Thermal Properties of Loaded Metal–Organic Frameworks: An Interplay of Quantum and Anharmonic Fluctuations*, *Journal of Chemical Theory and Computation* **15**, 3237–3249 (2019), [doi:10.1021/acs.jctc.8b01297](https://doi.org/10.1021/acs.jctc.8b01297).

- [2] A. H. Larsen *et al.*, *The Atomic Simulation Environment—a Python Library for Working with Atoms*, Journal of Physics: Condensed Matter **29**, 273002 (2017), [doi:10.1088/1361-648X/aa680e](https://doi.org/10.1088/1361-648X/aa680e).
- [3] A. Mazitov *et al.*, *PET-MAD as a Lightweight Universal Interatomic Potential for Advanced Materials Modeling*, Nature Communications **16**, 10653 (2025), [doi:10.1038/s41467-025-65662-7](https://doi.org/10.1038/s41467-025-65662-7).
- [4] S. N. Pozdnyakov and M. Ceriotti, *Smooth, Exact Rotational Symmetrization for Deep Learning on Point Clouds*, Advances in Neural Information Processing Systems **36** (2023), [conference paper](#).
- [5] L. Vanduyfhuys *et al.*, *QuickFF: A Program for a Quick and Easy Derivation of Force Fields for Metal–Organic Frameworks from Ab Initio Input*, Journal of Computational Chemistry **36**, 1015–1027 (2015).
- [6] M. Ceriotti, J. More, and D. E. Manolopoulos, *i-PI: A Python Interface for Ab Initio Path Integral Molecular Dynamics Simulations*, Computer Physics Communications **185**, 1019–1026 (2014).
- [7] G. J. Martyna, D. J. Tobias, and M. L. Klein, *Constant Pressure Molecular Dynamics Algorithms*, Journal of Chemical Physics **101**, 4177–4189 (1994).
- [8] G. J. Martyna, M. L. Klein, and M. Tuckerman, *Nosé–Hoover Chains: The Canonical Ensemble via Continuous Dynamics*, Journal of Chemical Physics **97**, 2635–2643 (1992).
- [9] M. Ceriotti, M. Parrinello, T. E. Markland, and D. E. Manolopoulos, *Efficient Stochastic Thermostatting of Path Integral Molecular Dynamics*, Journal of Chemical Physics **133**, 124104 (2010).
- [10] V. Kapil, J. Behler, and M. Ceriotti, *High Order Path Integrals Made Easy*, Journal of Chemical Physics **145**, 234103 (2016), [doi:10.1063/1.4971438](https://doi.org/10.1063/1.4971438).
- [11] M. F. Langer, M. Ceriotti, and collaborators, *SADMOF: Sparse Automatic Differentiation for Heat Capacity of Metal–Organic Frameworks*, project software and methodology under development (version used by this repository, 2026).